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(54) **SUBSTRATE DETACHING APPARATUS**

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H01J 37/32 (2006.01)
H01L 21/687 (2006.01)
H01L 23/467 (2006.01)

(52) **U.S. Cl.**

CPC .. ***H01L 21/67748*** (2013.01); ***H01J 37/32715*** (2013.01); ***H01L 21/68742*** (2013.01); ***H01L 23/467*** (2013.01)

(58) **Field of Classification Search**

CPC H01L 21/67748; H01L 21/68742; H01L 23/467; H01J 37/32715
See application file for complete search history.

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(57) **ABSTRACT**

A substrate detaching apparatus for detaching a substrate attracted to and held onto an attraction surface of an electrostatic chuck includes a moving unit configured to push and move the substrate in a direction parallel to the attraction surface.

10 Claims, 10 Drawing Sheets

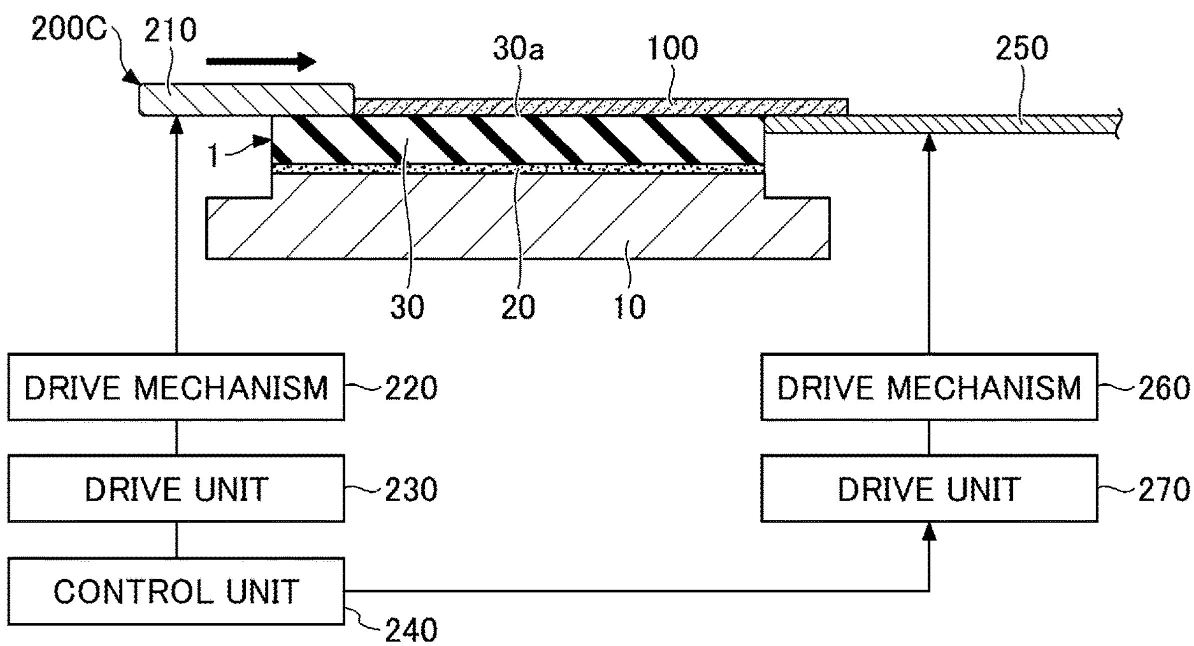


FIG.1A

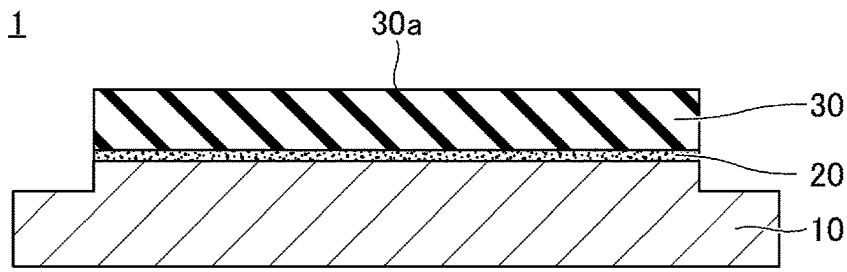


FIG.1B

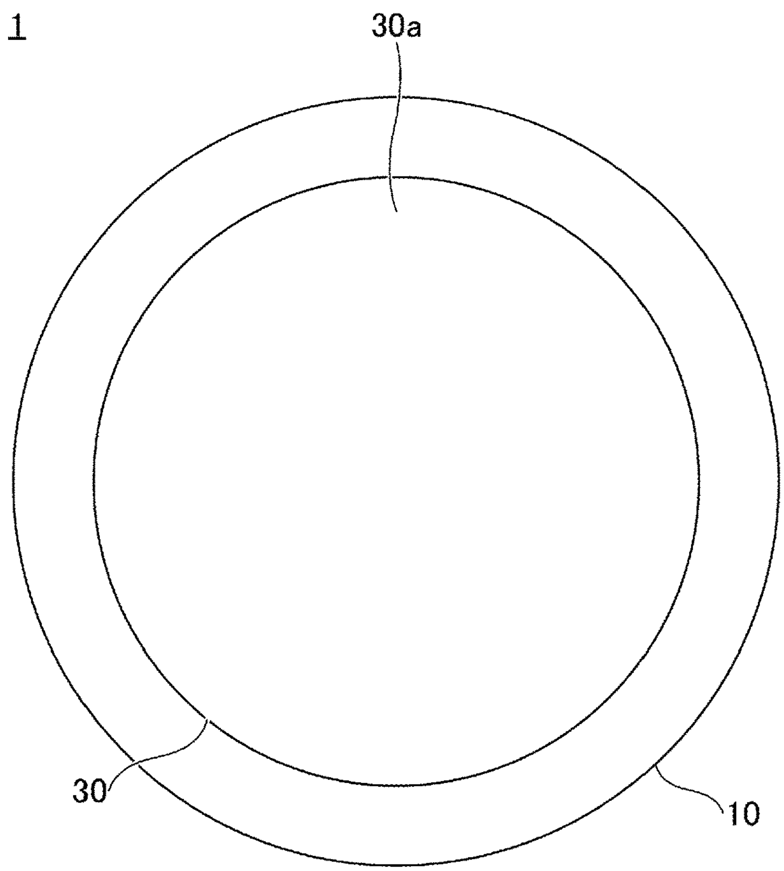


FIG.1C

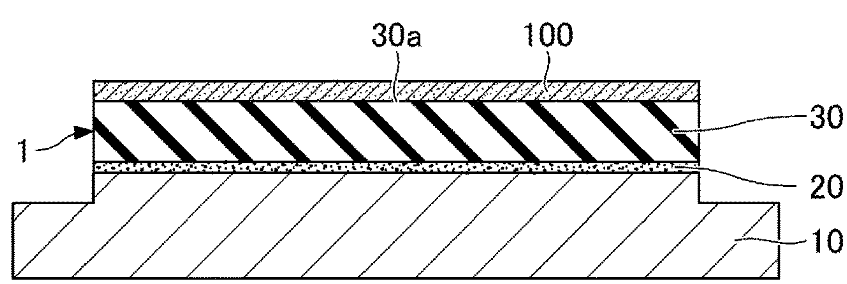


FIG.2A

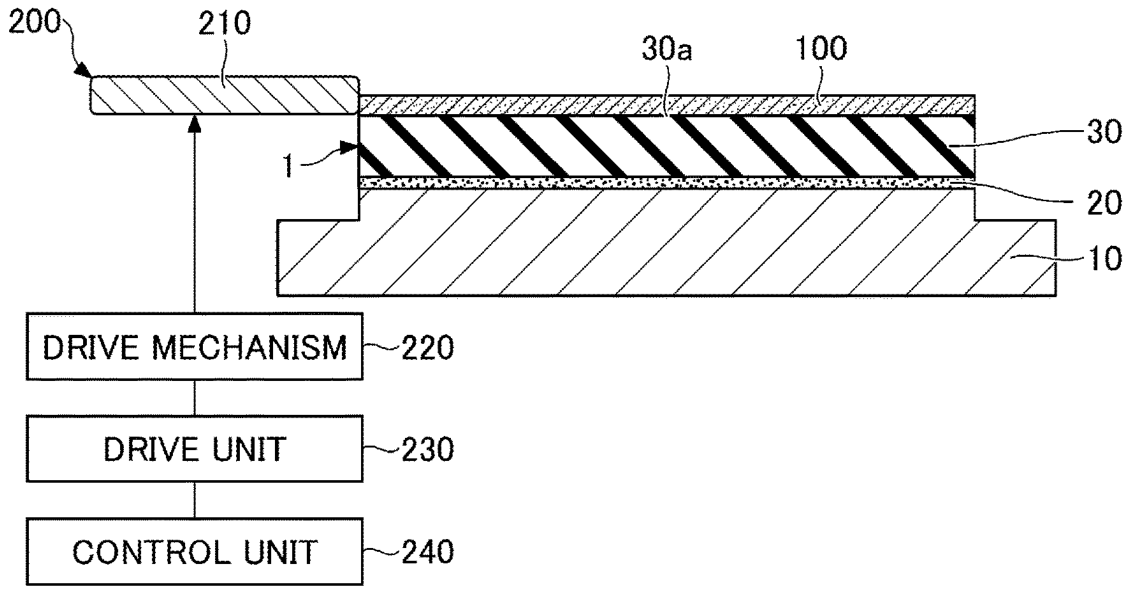

200
210
30a
100
30
1
20
10
DRIVE MECHANISM
220
DRIVE UNIT
230
CONTROL UNIT
240

FIG.2B

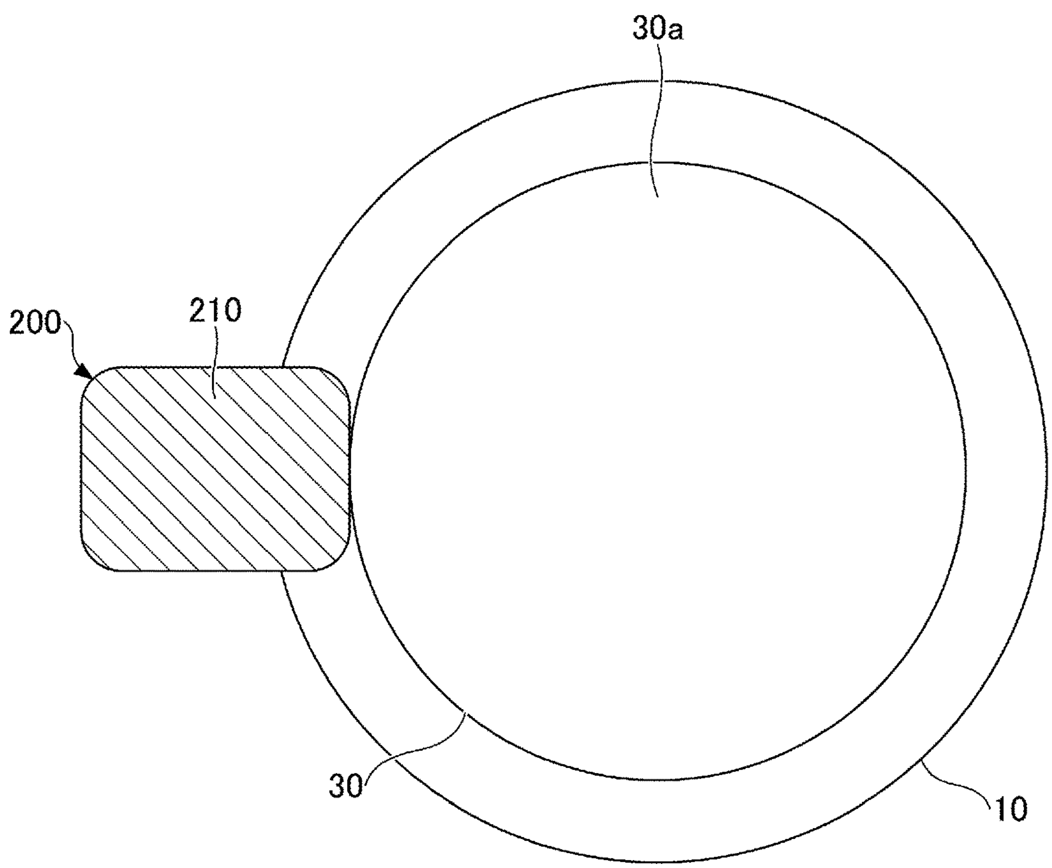

30a
200
210
30
10

FIG.3A

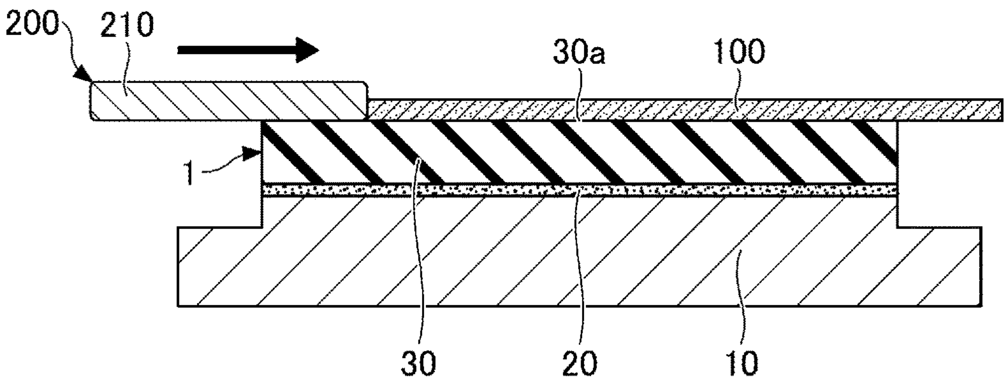


FIG.3B

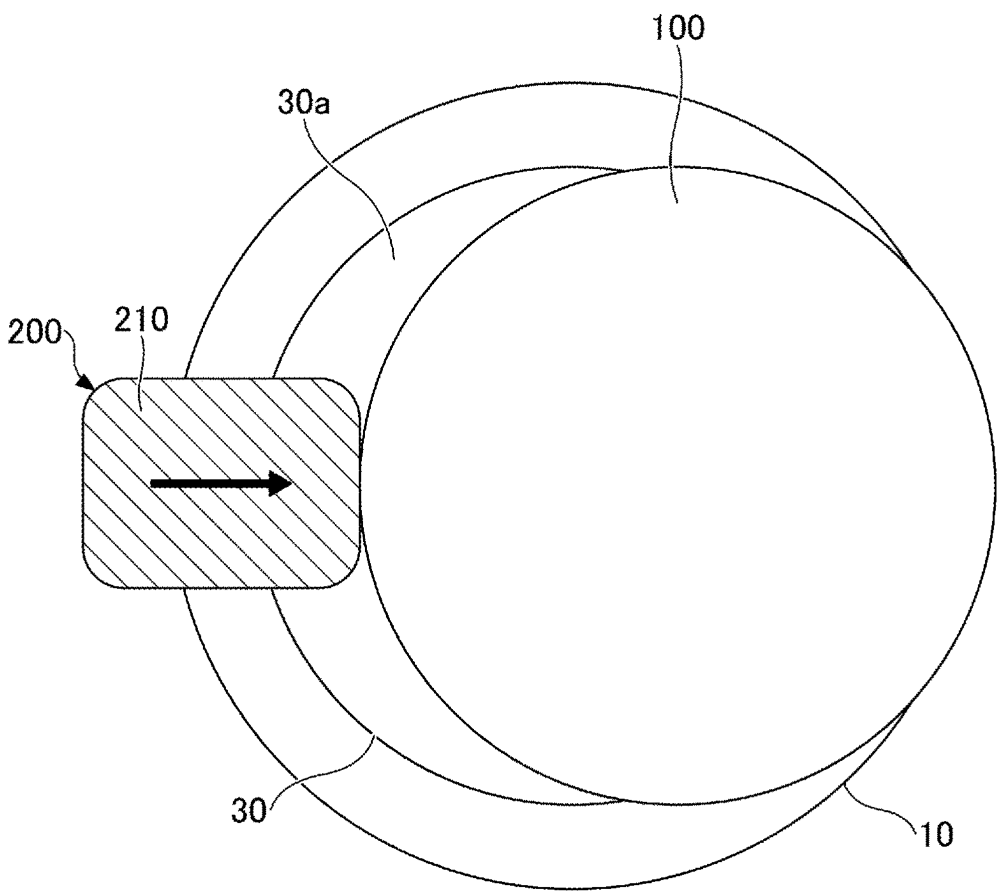


FIG.4

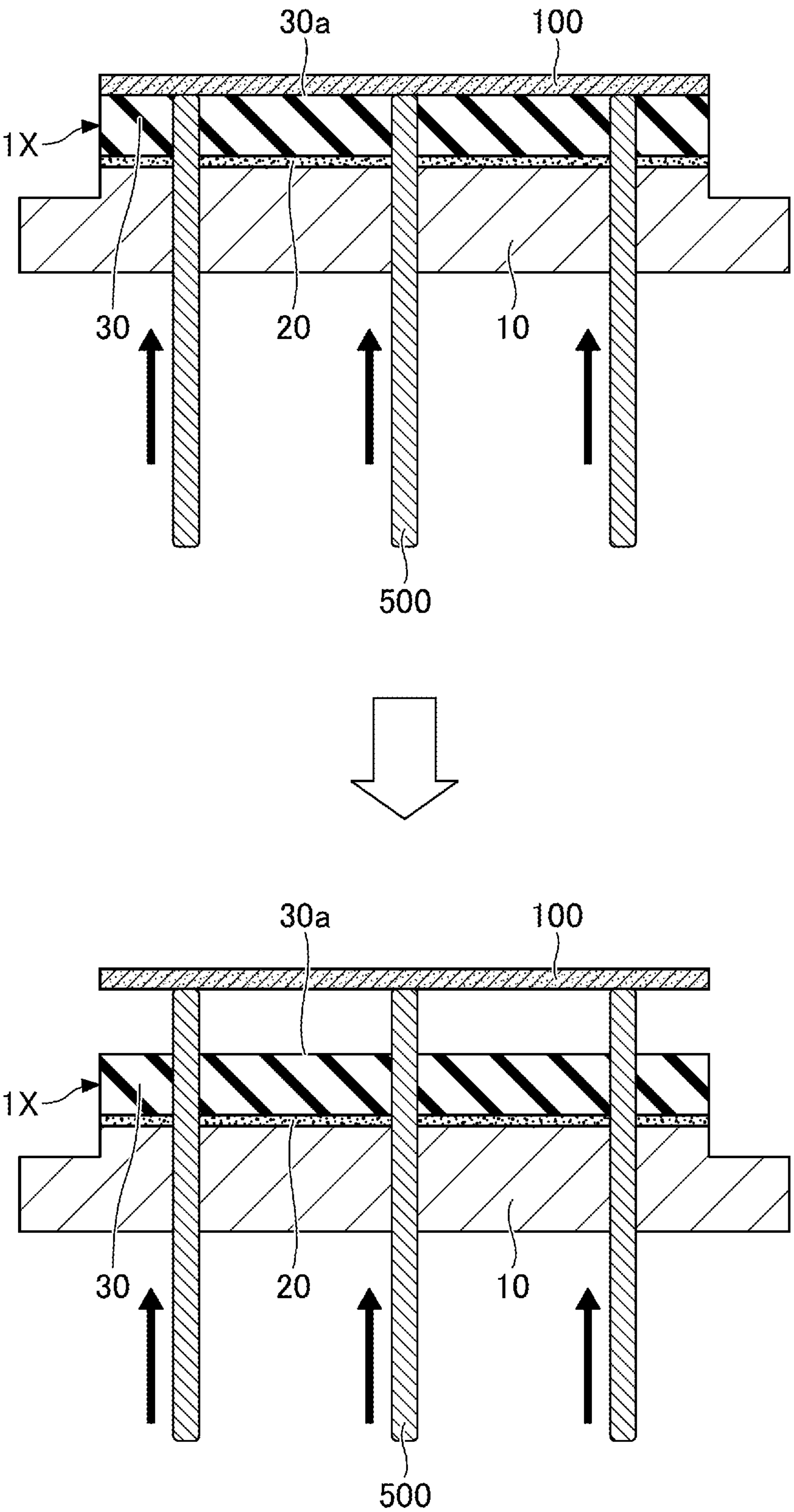


FIG.5A

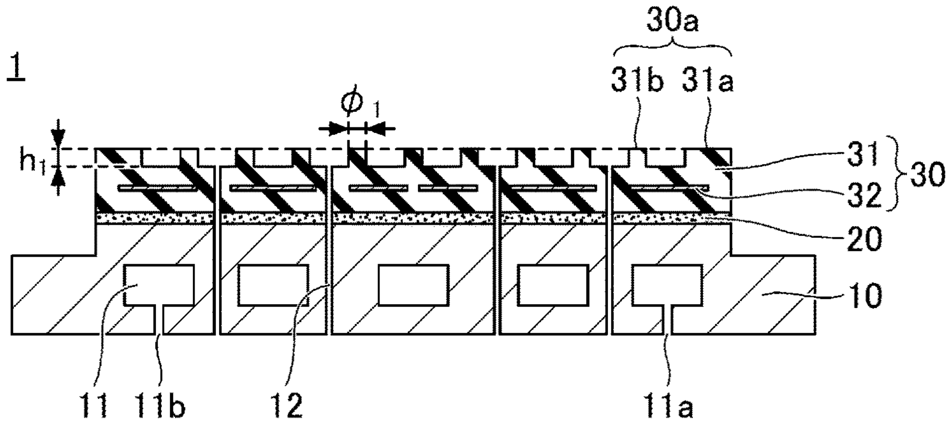

1
30a
31b 31a
ϕ_1
h_1
31
30
32
20
10
11 11b
12
11a

FIG.5B

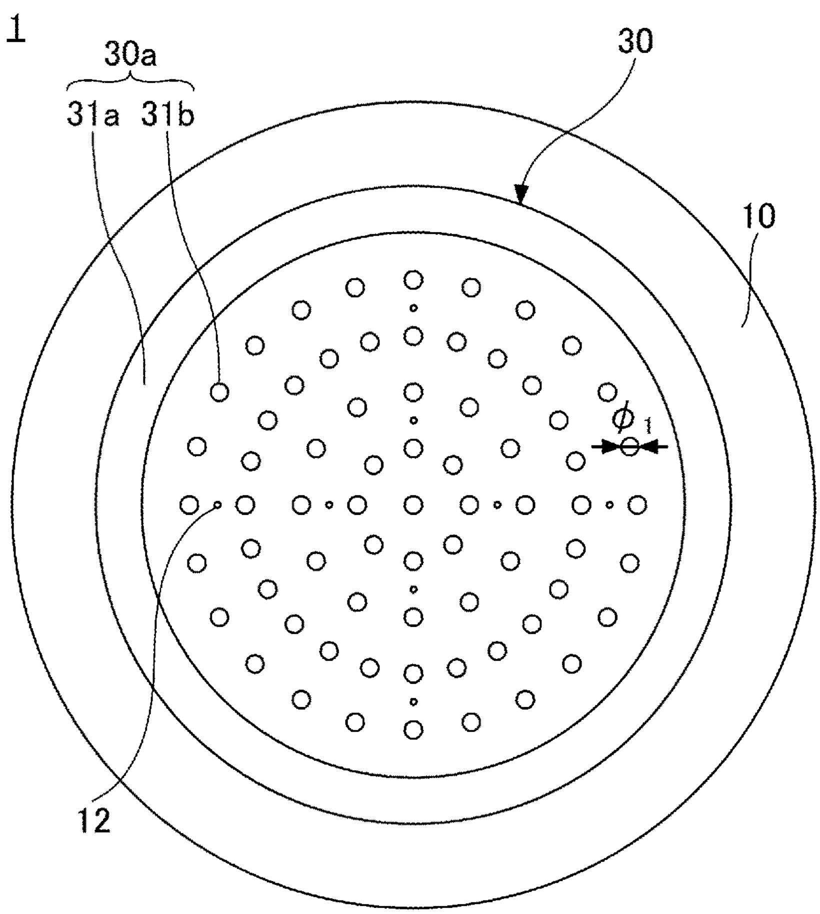

1
30a
31a 31b
30
10
ϕ_1
12

FIG.6

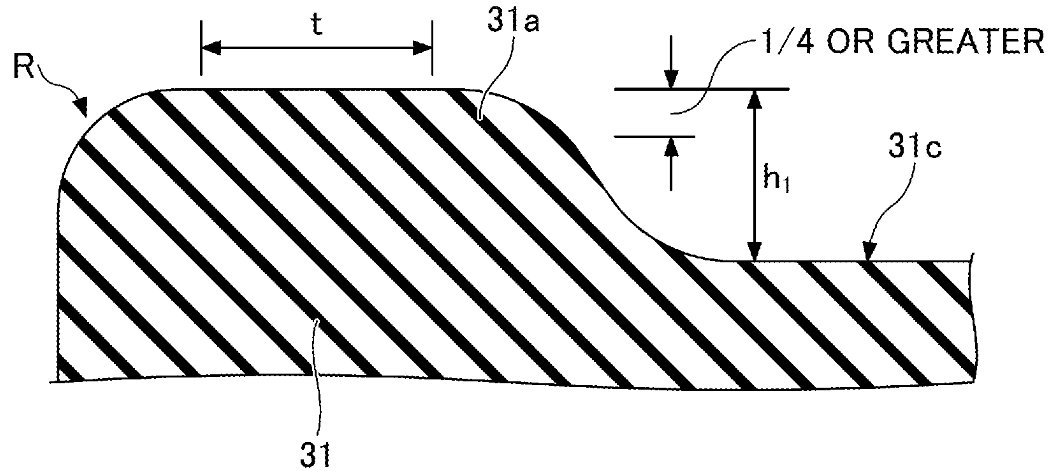


FIG.7

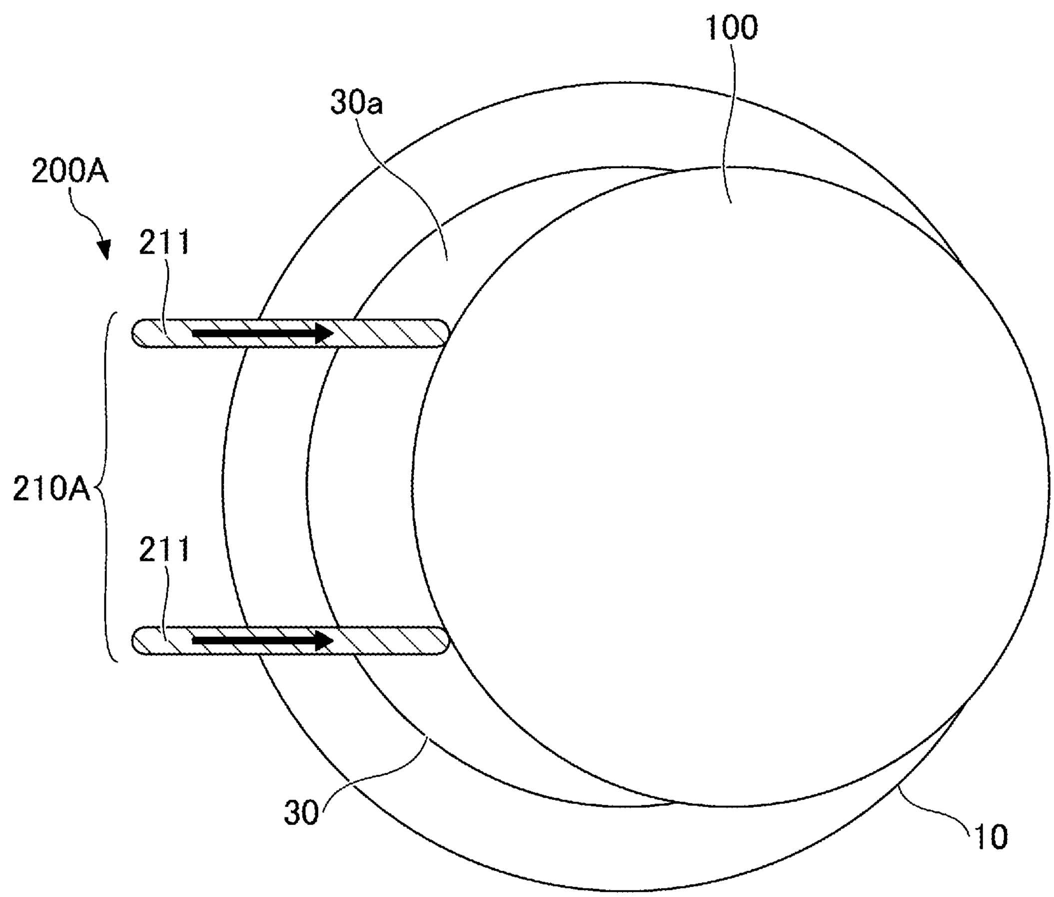


FIG.8

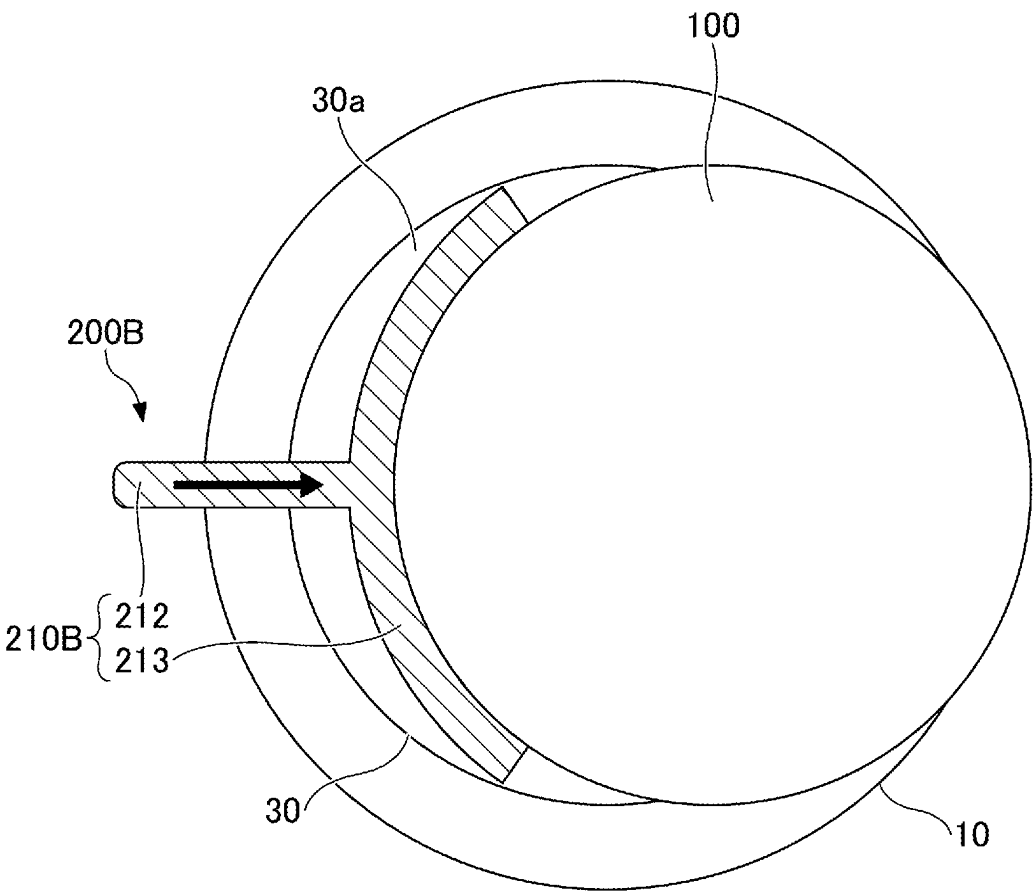


FIG.9A

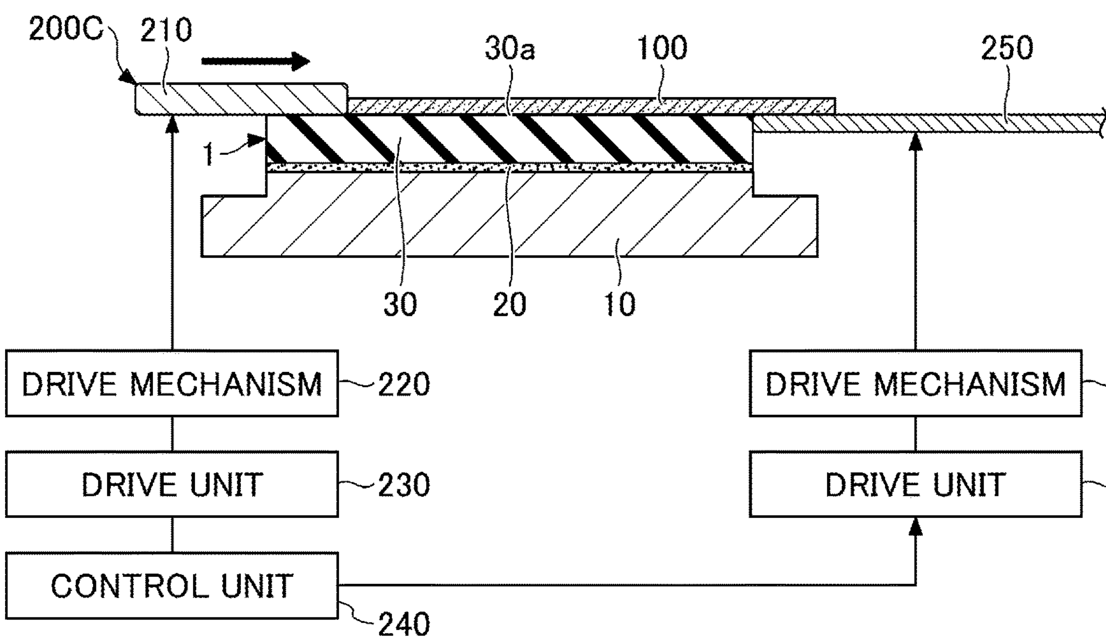

200C
210
30a
100
250
1
30
20
10
DRIVE MECHANISM
220
DRIVE UNIT
230
CONTROL UNIT
240
DRIVE MECHANISM
260
DRIVE UNIT
270

FIG.9B

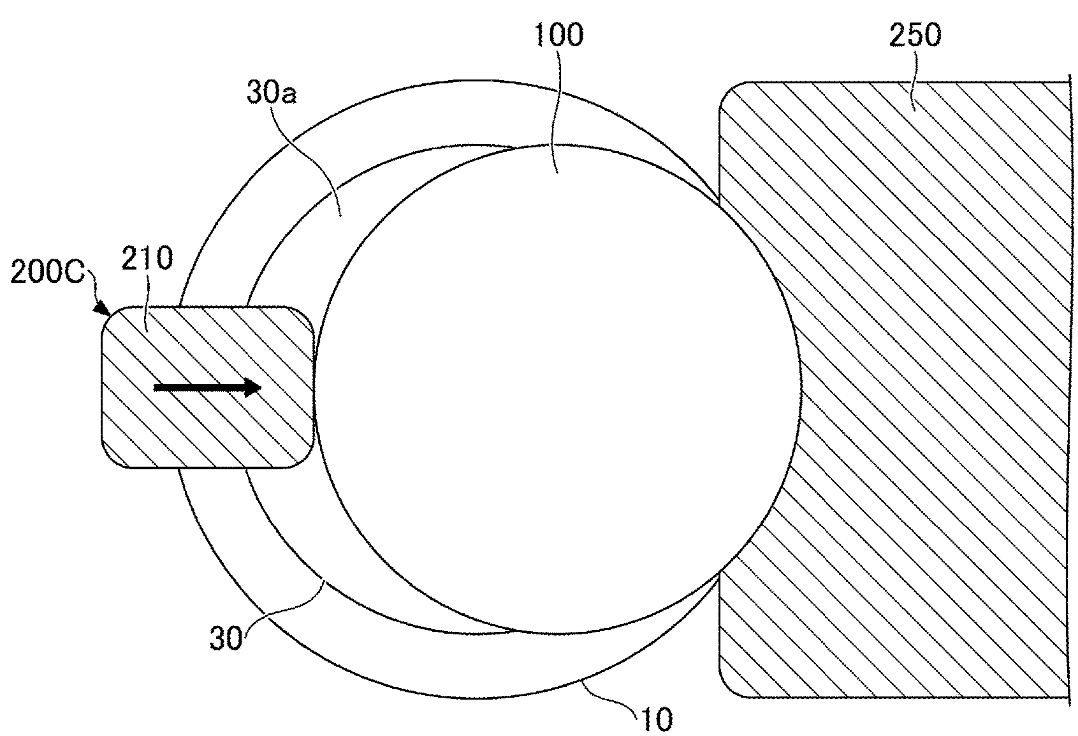

100
250
30a
200C
210
30
10

FIG.10A

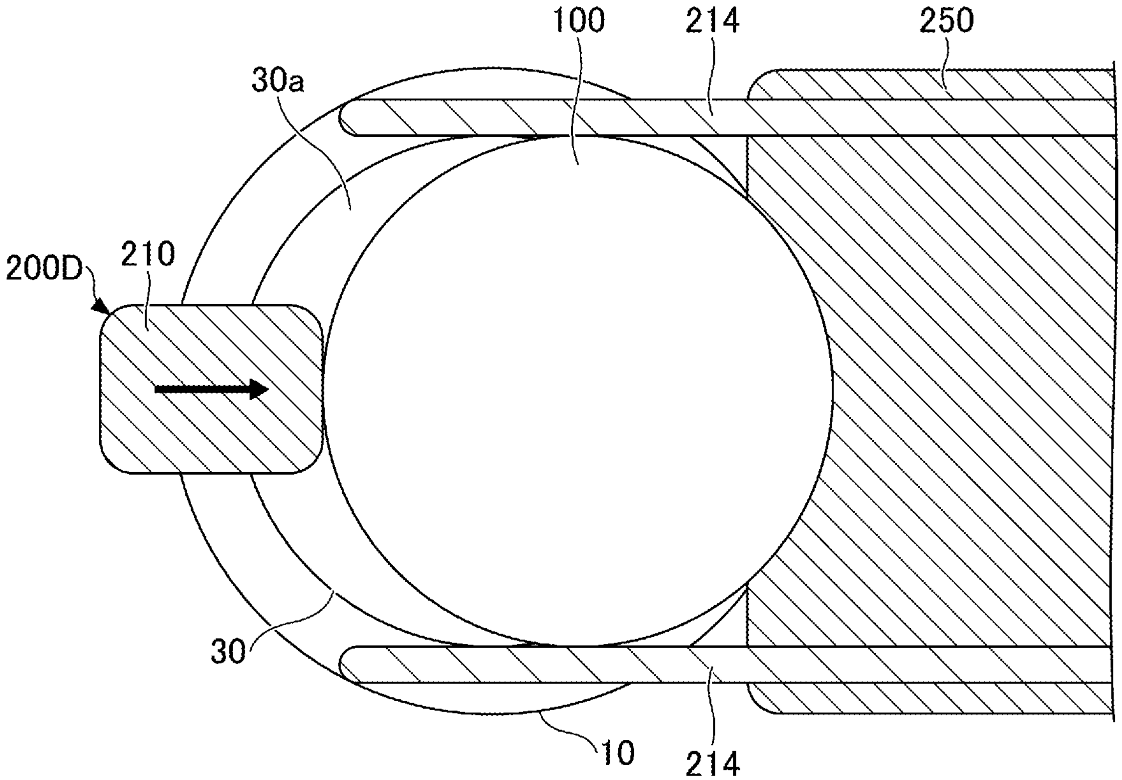


FIG.10B

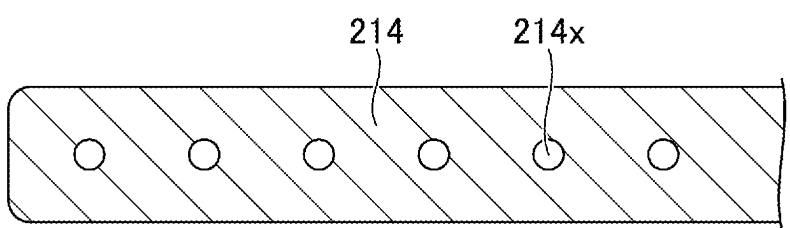
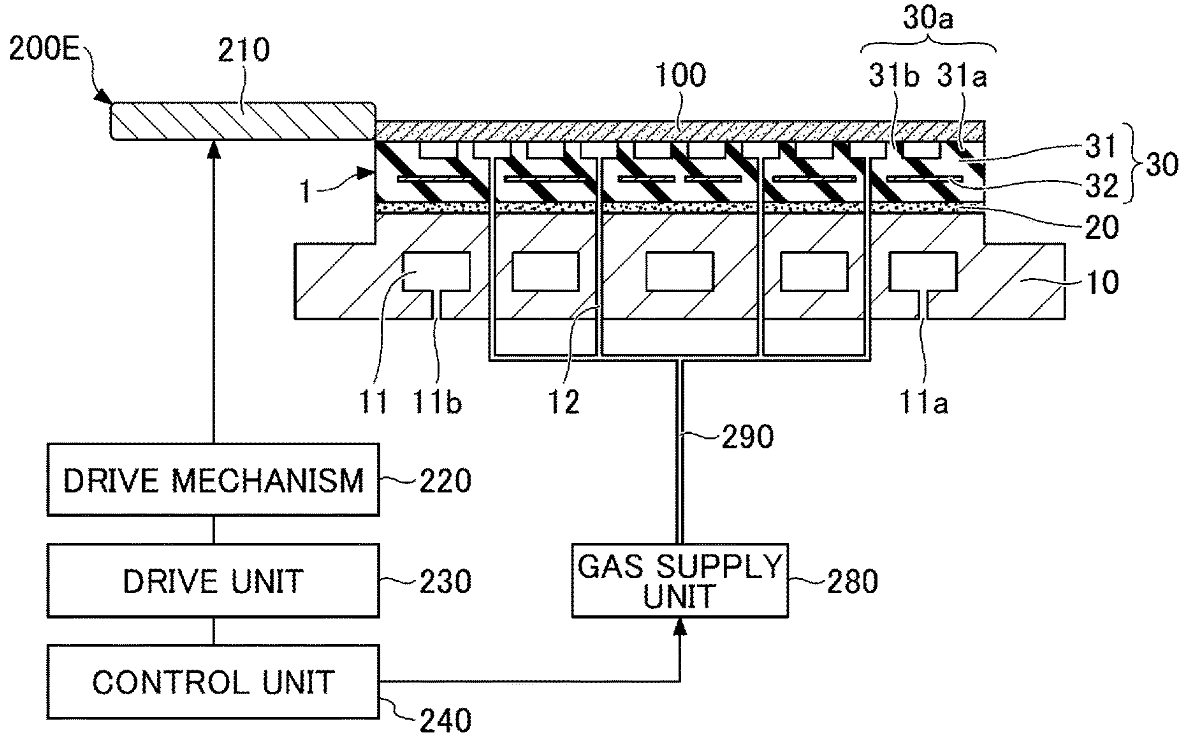


FIG.11


200E
210
100
30a
31b
31a
31
32
30
1
20
10
11
11b
12
11a
290
DRIVE MECHANISM
220
DRIVE UNIT
230
GAS SUPPLY UNIT
280
CONTROL UNIT
240

SUBSTRATE DETACHING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is based upon and claims priority to Japanese Patent Application No. 2021-069297 filed on Apr. 15, 2021, with the Japanese Patent Office, the entire contents of which are incorporated herein by reference.

FIELD

The disclosures herein relate to substrate detaching apparatuses.

BACKGROUND

A film deposition apparatus (e.g., a chemical vapor deposition apparatus, a physical vapor deposition apparatus, or the like) and a plasma etching apparatus are used in the process of manufacturing a semiconductor device such as an IC (integrated circuit) or an LSI (large scale integration). These apparatuses have a stage for holding a substrate in place with high accuracy in a vacuum processing chamber.

An example of such a stage is a substrate holding apparatus that attracts and holds a substrate, i.e., an object to be attracted, with an electrostatic chuck mounted on a baseplate. After the completion of processing of the substrate, lift pins that are movable perpendicularly to the attraction surface are used to vertically lift up and detach the substrate from the electrostatic chuck.

The speed and accuracy of a substrate detaching action vary due to the effect of a residual attraction force of the electrostatic chuck, which gives rise to a problem of a delay in the transfer of a substrate and misalignment upon detachment. Various studies have been conducted to reduce the effect of a residual attraction force of an electrostatic chuck, but a further reduction of the effect of a residual attraction force is required.

In consideration of the above, it may be desired to provide a substrate detaching apparatus that is capable of reducing the effect of a residual attraction force of an electrostatic chuck.

RELATED-ART DOCUMENTS

Patent Document

[Patent Document 1] Japanese Laid-open Patent Publication No. 2015-216391

SUMMARY

According to an aspect of the embodiment, a substrate detaching apparatus for detaching a substrate attracted to and held onto an attraction surface of an electrostatic chuck includes a moving unit configured to push and move the substrate in a direction parallel to the attraction surface.

The object and advantages of the embodiment will be realized and attained by means of the elements and combinations particularly pointed out in the claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention, as claimed.

BRIEF DESCRIPTION OF DRAWINGS

FIGS. **1**A through **1**C are drawings providing schematic illustrations of a substrate holding apparatus;

FIGS. **2**A and **2**B are drawings illustrating an example of a substrate detaching apparatus according to a first embodiment;

FIGS. **3**A and **3**B are drawings illustrating the functioning of the substrate detaching apparatus according to the first embodiment;

FIG. **4** is a cross-sectional view illustrating a related-art method of detaching a substrate from a substrate holding apparatus;

FIGS. **5**A and **5**B are drawings illustrating a more specific configuration of the substrate holding apparatus;

FIG. **6** is a drawing illustrating the shape of an annular protrusion constituting an attraction surface;

FIG. **7** is a plan view illustrating an example of a substrate detaching apparatus according to a first variation of the first embodiment;

FIG. **8** is a plan view illustrating an example of a substrate detaching apparatus according to a second variation of the first embodiment;

FIGS. **9**A and **9**B are drawings illustrating an example of a substrate detaching apparatus according to a third variation of the first embodiment;

FIGS. **10**A and **10**B are drawings illustrating an example of a substrate detaching apparatus according to a fourth variation of the first embodiment; and

FIG. **11** is a cross-sectional view illustrating an example of a substrate detaching apparatus according to a fifth variation of the first embodiment.

DESCRIPTION OF EMBODIMENTS

In the following, embodiments will be described by referring to the accompanying drawings. In these drawings, the same elements are referred to by the same references, and a duplicate description thereof may be omitted.

First Embodiment

The first embodiment is directed to a substrate detaching apparatus that is used together with a substrate holding apparatus having an electrostatic chuck and that detaches a substrate attracted to and held onto the surface of the electrostatic chuck. The general configuration of the substrate holding apparatus to which the substrate detaching apparatus is applied will be described first, and, then, the substrate detaching apparatus will be described.

[General Configuration of Substrate Holding Apparatus]

FIGS. **1**A through **1**C are drawings providing schematic illustrations of the substrate holding apparatus. FIG. **1**A and FIG. **1**C are cross-sectional views, and FIG. **1**B is a plan view. As illustrated in FIG. **1**A and FIG. **1**B, a substrate holding apparatus **1** includes a baseplate **10**, an adhesive layer **20**, and an electrostatic chuck **30** as main components.

The electrostatic chuck **30** is fixedly attached to one surface of the baseplate **10** through the adhesive layer **20**. A surface **30***a* is an attraction surface onto which an object is attracted and held. As illustrated in FIG. **1**C, the substrate holding apparatus **1** utilizes electrostatic charge to attract and hold a substrate **100** (e.g., a semiconductor wafer or the like), i.e., the object to be attracted, on the attraction surface **30***a* of the electrostatic chuck **30**.

[Substrate Detaching Apparatus]

FIGS. **2**A and **2**B are drawings illustrating an example of the substrate detaching apparatus according to the first embodiment. FIG. **2**A illustrates a cross-sectional view, and FIG. **2**B illustrates a plan view. In FIG. **2**B, a drive mechanism **220**, a drive unit **230**, and a control unit **240** illustrated in FIG. **2**A are omitted from illustration.

As illustrated in FIGS. **2**A and **2**B, a substrate detaching apparatus **200** detaches from the electrostatic chuck **30** the substrate **100** attracted and held by the electrostatic chuck **30** of the substrate holding apparatus **1**, and includes a moving unit **210**, the drive mechanism **220**, the drive unit **230**, and the control unit **240**. The substrate detaching apparatus **200** may be used together with the substrate holding apparatus **1** having the electrostatic chuck **30**.

The moving unit **210** serves to push the substrate **100** in a horizontal direction (i.e., a direction parallel to the attraction surface **30***a* of the electrostatic chuck **30**) to move the substrate **100**. The moving unit **210** is made of a metal, for example, and is supported by the drive mechanism **220** such as to be at least capable of reciprocating movement in the horizontal direction. Examples of the metal constituting the moving unit **210** include stainless steel, aluminum alloy, and the like. As long as the substrate **100** can be moved along the attraction surface **30***a* of the electrostatic chuck **30**, the moving unit **210** itself does not have to be arranged parallel to the attraction surface **30***a*. The moving unit **210** may be part of a robotic arm.

The drive mechanism **220** includes a drive force generating unit including a motor, and includes a drive force transmitting unit including a gear for transmitting the drive force generated by the drive force generating unit to the moving unit **210**, for example. The drive unit **230** includes, for example, a drive circuit for supplying electrical power to the drive force generating unit of the drive mechanism **220**, and operates in response to commands from the control unit **240**. The control unit **240** issues commands to the drive unit **230** to control the position of the moving unit **210**, for example. The moving unit **210** may further be configured to move in a vertical direction or in a rotational direction, in addition to required reciprocating movement in the horizontal direction.

The control unit **240** may be configured to include a CPU (central processing unit), a ROM (read only memory), a main memory, and the like, for example. In this case, various functions of the control unit **240** may be implemented by the CPU executing programs that are loaded from the ROM or the like to the main memory. A sensor for detecting the position of the substrate **100** may be disposed around the substrate **100**, and the control unit **240** may control the moving unit **210** based on information from the sensor. The control unit **240** may include, as needed, other components, such as an interface for transmitting and receiving information to and from an external unit.

The moving unit **210** stays on standby at a position away from the substrate holding apparatus **1**, and is then moved to a position from which the side edge of the substrate **100** can be pushed as illustrated in FIG. **2** after the completion of plasma processing or the like of the substrate **100** attracted and held by the substrate holding apparatus **1**. The lower surface of the moving unit **210** may be positioned on the same plane as the attraction surface **30***a*, for example, but may be positioned above the attraction surface **30***a*.

FIGS. **3**A and **3**B are drawings illustrating the functioning of the substrate detaching apparatus according to the first embodiment. FIG. **3**A illustrates a cross-sectional view, and FIG. **3**B illustrates a plan view. In FIGS. **3**A and **3**B, the drive mechanism **220**, the drive unit **230**, and the control unit **240** illustrated in FIG. **2**A are omitted from illustration. In FIGS. **3**A and **3**B, the moving unit **210** of the substrate detaching apparatus **200** moves in the direction indicated by an arrow while pressing the substrate **100**, so that the substrate **100** moves horizontally on the attraction surface **30***a* of the electrostatic chuck **30**.

The substrate **100** attracted and held onto the attraction surface **30***a* of the substrate holding apparatus **1** is pushed in the direction indicated by the arrow by the moving unit **210** as illustrated in FIG. **3** after stopping the application of voltage for generating electrostatic charge, and is thus moved horizontally on the attraction surface **30***a* of the electrostatic chuck **30**. The movement illustrated in FIG. **3** continues until the entirety of the substrate **100** is moved out of the attraction surface **30***a* of the electrostatic chuck **30**.

The substrate **100** detached from the attraction surface **30***a* of the electrostatic chuck **30** is placed on a loading arm, for example, for transfer to the next processing site. Thereafter, the moving unit **210** moves to a position away from the substrate holding apparatus **1** to stay on standby. A next substrate is then attracted and held onto the attraction surface **30***a* of the electrostatic chuck **30**, and is processed by plasma or the like. Subsequently, the moving unit **210** is moved to the position illustrated in FIG. **2**. The next substrate is detached from the attraction surface **30***a* in the same manner as illustrated in FIG. **3**, followed by being placed on the loading arm to be transferred to the next processing site. This cycle is repeated as many times as needed.

As described above, the substrate detaching apparatus **200** moves the substrate **100** electrostatically attracted and held onto the attraction surface **30***a* of the electrostatic chuck **30** in a direction parallel to the attraction surface **30***a* to detach it.

FIG. **4** is a cross-sectional view illustrating a related-art method of detaching a substrate from a substrate holding apparatus. A substrate holding apparatus **1**X has through holes into which a plurality of lift pins **500** are inserted. The substrate **100** attracted and held onto the attraction surface **30***a* of the substrate holding apparatus **1**X is lifted up and detached by the lift pins **500**, which push the lower surface of the substrate **100** by moving in the direction indicated by arrows (i.e., in the vertical direction) after stopping the application of voltage for generating electrostatic charge.

However, in the method illustrated in FIG. **4**, it takes time for a residual attraction force to disappear after the stoppage of the application of voltage. A waiting time is thus needed, thereby resulting in an increase in the processing time in some cases. In other cases, raising the lift pins **500** after the passage of a fixed time period following the stoppage of the application of voltage may create a misalignment of the substrate **100** due to the uneven distribution of the residual attraction force. This may prevent the normal transfer of the substrate **100**.

In contrast, the method illustrated in FIG. **3** improves the above-noted problems by moving the substrate **100** electrostatically attracted and held onto the attraction surface **30***a* of the electrostatic chuck **30** in a direction parallel to the attraction surface **30***a*. This will be described in detail in the following.

In the related-art method illustrated in FIG. **4**, vertically detaching the substrate **100** attracted and held onto the attraction surface **30***a* requires that a force stronger than a residual attraction force be evenly applied to the substrate **100**. In contrast, the method according to the first embodiment illustrated in FIG. **3** can detach the substrate **100** by applying, in a direction parallel to the attraction surface **30***a*, a force exceeding the maximum static friction force generated between the attraction surface **30***a* and the substrate **100**.

F' may denote a maximum static friction force, and p may denote the coefficient of maximum static friction, with N being a normal force. In such a case, the following relationship is satisfied: F'=μN. The normal force N is the sum of the mass of the substrate **100** and a residual attraction force. The maximum static friction coefficient μ depends on the state of materials in contact with each other (e.g., the property of material, surface roughness, etc.), but is generally less than or equal to 1 (approximately 0.3 to 0.6). As a result, the substrate **100** can be moved horizontally by applying a force that is lower than the force required to lift up the substrate **100** vertically against the residual attraction force.

Further, the method of lifting up the substrate **100** in the direction perpendicular to the attraction surface **30***a* with the lift pins **500** may readily create elastic deformation of the substrate **100** due to the residual attraction force acting against the upward movement of the lift pins **500** and also a misalignment of the substrate **100** in plane coordinates upon recovery from the elastic deformation. In contrast, the method of horizontally moving the substrate **100** does not cause deformation of the substrate **100**, and thus does not cause a resultant misalignment of the substrate **100** in plane coordinates.

As is described above, the method of horizontally moving the substrate **100** can detach the substrate **100** from the attraction surface **30***a* with a lower force than in the related-art method of vertically moving the substrate **100**. Further, the misalignment of the substrate **100** can be reduced at the time of detaching the substrate **100**, which enables the normal transfer of the substrate **100**. Since the time required to wait for the disappearance of a residual attraction force upon stopping the application of voltage can be shortened, the processing time can also be shortened.

[Specific Example of Substrate Holding Apparatus]

A specific example of the substrate holding apparatus to which the substrate detaching apparatus is applicable will be described in the following. It may be noted that the following description is intended to illustrate only an example of the substrate holding apparatus to which the substrate detaching apparatus is applicable, and is not intended to provide a limiting example.

FIGS. **5**A and **5**B are drawings illustrating a more specific configuration of the substrate holding apparatus. FIG. **5**A is a cross-sectional view, and FIG. **5**B is a plan view. As illustrated in FIG. **5**A and FIG. **5**B, the substrate holding apparatus **1** includes the baseplate **10**, the adhesive layer **20**, and the electrostatic chuck **30** as main components.

The baseplate **10** serves to support the electrostatic chuck **30**. The thickness of the baseplate **10** may approximately be in a range of 20 mm to 40 mm, for example. The baseplate **10** may be made of a metallic material such as aluminum or a cemented carbide, or a composite material of the noted metallic material and ceramics, for example, and may serve as an electrode or the like for controlling plasma. From the viewpoint of availability, ease of processing, good thermal conductivity, and the like, aluminum or an alloy thereof may preferably be used with the surface thereof being subjected to an alumite treatment (i.e., formation of an insulating layer), for example.

Supplying a predetermined high-frequency electric power to the baseplate **10** enables the control of energy with which generated plasma ions impact a substrate held on the electrostatic chuck **30**, thereby achieving an efficient etching process, for example.

The baseplate **10** may have a water pathway **11** disposed therein. The water pathway **11** has a coolant-water inlet **11***a* at one end and a coolant-water outlet **11***b* at the other end. The water pathway **11** is connected to a coolant water control apparatus (not shown) provided outside the substrate holding apparatus **1**. The coolant water control apparatus (not shown) feeds coolant water into the water pathway **11** at the coolant-water inlet **11***a*, and receives coolant water discharged from the coolant-water outlet **11***b*. Circulating coolant water through the water pathway **11** to cool the baseplate **10** causes a substrate held on the electrostatic chuck **30** to be cooled.

The electrostatic chuck **30** is fixedly attached to one surface of the baseplate **10** through the adhesive layer **20**. A silicone-based adhesive, for example, may be used as the adhesive layer **20**. The thickness of the adhesive layer **20** is approximately in a rage of 0.01 mm to 1.0 mm, for example. The adhesive layer **20** serves to bond the baseplate **10** to the electrostatic chuck **30** and to reduce the stress caused by the difference in thermal expansion coefficients between the ceramic electrostatic chuck and the aluminum baseplate **10**. It may be noted that the electrostatic chuck **30** may alternatively be secured to the baseplate **10** with screws.

The electrostatic chuck **30** includes a base **31** and electrostatic electrodes **32** as main components. The plane shape of the electrostatic chuck **30**, which is determined in accordance with the shape of a substrate, may be circular, for example. The diameter of a substrate that is an object to be attracted by the electrostatic chuck **30** may be 8, 12, or, 18 inches, for example. The electrostatic chuck **30** may be a Johnsen-Rahbeck electrostatic chuck or a Coulomb-type electrostatic chuck.

A plan view refers to a view of an object taken in the direction perpendicular to the attraction surface **30***a* of the base **31**, and a plane shape refers to the shape of an object as viewed in the direction perpendicular to the attraction surface **30***a* of the base **31**.

The base **31** is made of a dielectric material, which may be a ceramic such as aluminum oxide (Al_2O_3), aluminum nitride (AlN), or the like. The thickness of the base **31** may approximately be in a range of 0.5 mm to 10 mm. The relative permittivity of the base **31** at 1 KHz may approximately be in a range of 9 to 10.

The electrostatic electrode **32** is a thin-film electrode that is embedded in the base **31**. The electrostatic electrodes **32** are coupled to a power supply (not shown) provided separately from the substrate holding apparatus **1**. Upon receiving a predetermined voltage from the power supply, the electrostatic electrodes **32** generate an electrostatic attracting force with respect to a substrate. This causes the substrate to be attracted to and held onto the attraction surface **30***a* of the base **31** of the electrostatic chuck **30**. The attracting force increases as the voltage applied to the electrostatic electrode **32** increases. The electrostatic electrode **32** may have either a monopole structure or a dipole structure. Tungsten, molybdenum, or the like may be used as the material of the electrostatic electrode **32**.

A heating element may be provided inside the base **31** to generate heat by receiving voltage from outside the substrate holding apparatus **1** thereby to increase the temperature of the attraction surface **30***a* of the base **31** to a predetermined temperature.

The periphery of the upper surface of the base **31** is provided with an annular protrusion **31***a*, for example, in the form of a ring in a plan view. A number of columnar protrusions **31***b* having a cylindrical shape or the like are arranged in a polka-dot pattern inside the annular protrusion **31***a* in a plan view. The height h**1** of the top surface is the same for the annular protrusion **31***a* and the columnar protrusions **31***b*. The height h**1** may be, for example, in a range of 10 μm to 30 μm. The diameter ϕ**1** of the top surface of the columnar protrusions **31***b* may be, for example, in a range of 1.0 mm to 2.0 mm. The top surface of the annular protrusion **31***a* and the top surface of the columnar protrusions **31***b* constitute the attraction surface **30***a* which attracts and holds an object.

The electrostatic chuck **30** and the baseplate **10** have gas pathways **12** provided therein that supply a gas for cooling a substrate attracted to and held onto the electrostatic chuck **30**. The gas pathways **12** are holes formed in the baseplate **10**, the adhesive layer **20**, and the base **31**. The number of gas pathways **12**, which can properly be determined according to need, may approximately be ten to one hundred, for example. An inert gas, for example, is introduced into the gas pathways **12** from a source situated outside the substrate holding apparatus **1**. Helium gas, argon gas, or the like, for example, may be used as the inert gas.

The effect of a residual attraction force of the electrostatic chuck **30** is preferably reduced when the substrate **100** is moved horizontally and detached from the attraction surface **30***a* of the electrostatic chuck **30**. The effect of a residual attraction force of the electrostatic chuck **30** is effectively reduced by reducing the static friction coefficient of the attraction surface **30***a*. In order to reduce the static friction coefficient of the attraction surface **30***a*, the smaller the surface roughness of the attraction surface **30***a* is, the better the outcome will be. Namely, it is preferable to reduce as much as possible the roughness of the upper surface of the annular protrusion **31***a* and the upper surface of the columnar protrusions **31***b* that serve as the attraction surface **30***a*. For example, the surface roughness Ra of the upper surface of the annular protrusion **31***a* and the upper surface of the columnar protrusions **31***b* that serve as the attraction surface **30***a* may be greater than or equal to 0 and less than or equal to 0.2 μm.

The material of the base **31** of the electrostatic chuck **30** is often a ceramic. For the purpose of reducing the surface roughness of the attraction surface **30***a*, a denser structure having a porosity of less than or equal to 1% is preferable. The surface roughness of the attraction surface **30***a* also greatly depends the type of material used for the base **31**. Using a ceramic such as hexagonal boron nitride is expected to achieve a static friction coefficient that is far lower than 0.1. However, some ceramic materials, such as aluminum-oxide-based ceramics, are generally chosen as the material for the base **31** because of the trade-offs with other requirements for the material of an electrostatic chuck (e.g., plasma resistance, mechanical strength, hardness, price, etc.).

Further, when the substrate **100** is horizontally moved and detached from the attraction surface **30***a* of the electrostatic chuck **30**, the sliding surfaces are constituted by the substrate **100** and the attraction surface **30***a*. Because of this, there is a concern about the generation of particles due to sliding (i.e., the generation of fragments due to wearing from friction). In consideration of the size reduction and density increase of semiconductor circuits, the generation of particles needs be strictly controlled. Reducing the generation of particles requires reducing sliding resistance as much as possible. Improving the smoothness of the attraction surface **30***a* to reduce the static friction coefficient of the attraction surface **30***a* is an effective measure in this case also.

The portion where particles are likely to be generated due to sliding movement of the substrate **100** and the attraction surface **30***a* is the periphery of the attraction surface **30***a*. It is thus preferable that the edges of the annular protrusion **31***a* situated at the periphery of the attraction surface **30***a* have an obtuse angle, and more preferably be a smooth curved surface. For example, as illustrated in FIG. **6**, the edges of the upper surface (i.e., the surface in contact with the substrate **100**) of the annular protrusion **31***a* constituting the attraction surface **30***a* preferably have a rounded profile.

Specifically, the annular protrusion **31***a* of the electrostatic chuck **30** preferably has a radius of curvature R greater than or equal to approximately 0.01 mm at an edge where the upper surface and the lateral surface intersect. Use of the radius of curvature R below 0.01 mm results in an increased sharpness at the edges and thus an increase in the likelihood of particle generation. De-chuck characteristics are also lowered. Such characteristic changes also depend on the size of the portion having the radius of curvature R. According to the present inventor's understanding, the portion having the radius of curvature R preferably accounts for approximately greater than or equal to a quarter of the height h**1** of the annular protrusion **31***a*. The size of this portion falling below the quarter results in an insufficient effect of the radius of curvature R and thus an increase in the likelihood of particle generation. De-chuck characteristics are also lowered.

The top surface of the annular protrusion **31***a* has, around the center thereof (i.e., the portion indicated by t in FIG. **6**), a mirror finish surface that is protected and sustained by a masking means. This arrangement also contributes to the prevention of particle generation and the improvement of de-chuck characteristics. Preferably, the surface roughness Ra of the mirror finish surface is about less than or equal to 0.2 μm. The surface **31***c* of the electrostatic chuck **30**, except for the annular protrusion **31***a*, is a blasted surface, and the surface roughness Ra thereof is typically in a range of approximately 0.2 μm to 1.0 μm. However, if additional blasting is performed to provide the radius of curvature R, the surface roughness Ra may further be reduced to, for example, 0.3 μm or less.

Various methods may enable the provision of the radius of curvature R to the annular protrusion **31***a*. For example, the annular protrusion **31***a* may be formed on the surface of the electrostatic chuck **30** by embossing, and, then, the edges of the annular protrusion **31***a* may be smoothed by post-processing such as polishing or blasting. Alternatively, the edges of the annular protrusion **31***a* may be smoothed at the time of forming the annular protrusion **31***a* on the surface of the electrostatic chuck **30** by embossing.

The above-noted arrangement has been directed to the annular protrusion **31***a*. Similarly, the edges of the upper surface (i.e., the surface in contact with the substrate **100**) of the columnar protrusions **31***b* serving as the attraction surface **30***a* preferably have a rounded profile. This can reduce the surface roughness of the attraction surface **30***a*.

Variations of First Embodiment

The variations of the first embodiment are directed to an example of a substrate detaching apparatus having a different configuration than that of the first embodiment. In connection with the variations of the first embodiment, a description of the same or similar constituent elements as those of the previously provided descriptions may be omitted as appropriate.

FIG. **7** is a plan view illustrating an example of a substrate detaching apparatus according to a first variation of the first embodiment. In FIG. **7**, the drive mechanism **220**, the drive unit **230**, and the control unit **240** illustrated in FIG. **2**A are omitted from illustration. In FIG. **7**, a moving unit **210**A of a substrate detaching apparatus **200**A moves in the direction indicated by arrows while pressing the substrate **100**, so that the substrate **100** moves horizontally on the attraction surface **30***a* of the electrostatic chuck **30**.

The substrate detaching apparatus **200**A may be structured such that, as illustrated in FIG. **7**, the moving unit **210**A has two or more elongate parts **211** spaced apart from each other and having the longitudinal direction thereof aligned in the direction of movement. The structure of the moving unit **210**A as illustrated in FIG. **7** can reduce a rotational displacement of the substrate **100** when the substrate **100** is moved in the direction indicated by the arrows. FIG. **7** illustrates an example having two elongate parts **211**. Similarly, a configuration having three or more elongate parts **211** has the same effects.

FIG. **8** is a plan view illustrating an example of a substrate detaching apparatus according to a second variation of the first embodiment. In FIG. **8**, the drive mechanism **220**, the drive unit **230**, and the control unit **240** illustrated in FIG. **2**A are omitted from illustration. In FIG. **8**, a moving unit **210**B of a substrate detaching apparatus **200**B moves in the direction indicated by arrows while pressing the substrate **100**, so that the substrate **100** moves horizontally on the attraction surface **30***a* of the electrostatic chuck **30**.

The substrate detaching apparatus **200**B may be structured such that, as illustrated in FIG. **8**, the moving unit **210**B has an elongate part **212** and a curved part **213** disposed at one end of the elongate part **212**. The curved part **213** has the same radius of curvature as the substrate **100**, and is capable of contacting with less than or equal to half of the circumference of the substrate **100** in a plan view. The structure of the moving unit **210**B as illustrated in FIG. **8** can reduce a rotational displacement of the substrate **100** when the substrate **100** is moved in the direction indicated by the arrows.

FIGS. **9**A and **9**B are drawings illustrating an example of the substrate detaching apparatus according to a third variation of the first embodiment. FIG. **9**A illustrates a cross-sectional view, and FIG. **9**B illustrates a plan view. In FIG. **9**B, the drive mechanisms **220** and **260**, the drive units **230** and **270**, and the control unit **240** illustrated in FIG. **9**A are omitted from illustration. In FIGS. **9**A and **9**B, the moving unit **210** of a substrate detaching apparatus **200**C moves in the direction indicated by an arrow while pressing the substrate **100**, so that the substrate **100** moves horizontally on the attraction surface **30***a* of the electrostatic chuck **30** toward a loading arm **250**.

As illustrated in FIGS. **9**A and **9**B, the substrate detaching apparatus **200**C differs from the substrate detaching apparatus **200** (see FIGS. **1**A through **1**C and the like) in that the loading arm **250**, a drive mechanism **260**, and a drive unit **270** are provided.

The loading arm **250** may be disposed at a position opposite from the moving unit **210** across the electrostatic chuck **30**. The loading arm **250** carries and transfers the substrate **100** detached from the electrostatic chuck **30** to the next processing site. The loading arm **250** may be made of a metal, for example, and is supported by the drive mechanism **260** such as to be capable of moving in a predetermined direction. Examples of the metal constituting the loading arm **250** include stainless steel, aluminum alloy, and the like.

The drive mechanism **260** includes a drive force generating unit including a motor, and includes a drive force transmitting unit including a gear for transmitting the drive force generated by the drive force generating unit to the loading arm **250**, for example. The drive unit **270** includes, for example, a drive circuit for supplying electrical power to the drive force generating unit of the drive mechanism **260**, and operates in response to commands from the control unit **240**. The control unit **240** issues commands to the drive unit **270** to control the position of the loading arm **250**, for example.

The moving unit **210** and the loading arm **250** stay on standby at a position away from the substrate holding apparatus **1**, and are then moved to the positions illustrated in FIGS. **9**A and **9**B after the completion of plasma processing or the like of the substrate **100** attracted and held by the substrate holding apparatus **1**. The loading surface of the loading arm **250** is situated on the line on which the center of the substrate **100** moves, and is coplanar with the attraction surface **30***a*.

The loading arm may be provided separately from the substrate detaching apparatus, or may be part of the substrate detaching apparatus as illustrated in FIGS. **9**A and **9**B.

FIGS. **10**A and **10**B are drawings illustrating an example of the substrate detaching apparatus according to a fourth variation of the first embodiment. FIG. **10**A illustrates a plan view, and FIG. **10**B illustrates a side elevation view of a guide rail as viewed from the position of the substrate. In FIGS. **10**A and **10**B, the drive mechanisms **220** and **260**, the drive units **230** and **270**, and the control unit **240** illustrated in FIG. **9**A are omitted from illustration. In FIGS. **10**A and **10**B, the moving unit **210** of a substrate detaching apparatus **200**D moves in the direction indicated by an arrow while pressing the substrate **100**, so that the substrate **100** moves horizontally on the attraction surface **30***a* of the electrostatic chuck **30** toward the loading arm **250**.

As illustrated in FIGS. **10**A and **10**B, the substrate detaching apparatus **200**D differs from the substrate detaching apparatus **200**C (see FIGS. **9**A and **9**B) in that a pair of guide rails **214** is provided.

The guide rails **214** are spaced apart from each other, and have the longitudinal direction thereof aligned in the direction of movement of the substrate **100**, thereby guiding the movement of the substrate **100** from both sides. The guide rails **214** are made of, for example, a metal. Examples of the metal constituting the guide rails **214** include stainless steel, aluminum alloy, and the like.

Preferably, each guide rail **214** extends to a position overlapping the loading arm **250** in a plan view. The provision of the guide rails **214** can reduce a rotational displacement of the substrate **100** when the substrate **100** is moved in the direction indicated by the arrow. In FIGS. **9**A and **9**B, the moving unit **210**A or the moving unit **210**B may be used in place of the moving unit **210**.

The guide rails **214** stay on standby at a position away from the substrate holding apparatus **1**, and are then moved together with the loading arm **250**, for example, to the position illustrated in FIGS. **10**A and **10**B after the completion of plasma processing or the like of the substrate **100** attracted and held by the substrate holding apparatus **1**.

The lateral surface of at least one of the guide rails **214** situated toward the substrate **100** preferably has a plurality of vacuum suction holes **214***x* arrayed in the longitudinal direction thereof, as illustrated in FIG. **10**B. Connecting the vacuum suction holes **214***x* to a vacuum suction apparatus allows particles generated by the movement of the substrate **100** to be removed from the vicinity of the substrate **100** to the outside. A gas may be advantageously supplied through the gas pathways **12** (see FIGS. **5**A and **5**B) to the attraction surface **30***a* of the electrostatic chuck **30** for the purpose of controlling the direction of particle flow during vacuum suction.

It may be noted that a plurality of vacuum suction holes may similarly be provided, as in FIG. **10**B, on the lateral surface of the curved part **213** facing the substrate **100** illustrated in FIG. **8**. This arrangement also allows particles generated by the movement of the substrate **100** to be removed from the vicinity of the substrate **100** to the outside.

FIG. **11** is a cross-sectional view illustrating an example of a substrate detaching apparatus according to a fifth variation of the first embodiment. The substrate detaching apparatus **200**E illustrated in FIG. **11** is configured such that the control unit **240** can control a gas supply unit **280** disposed outside the substrate holding apparatus **1**.

As illustrated in FIG. **11**, the substrate detaching apparatus **200**E differs from the substrate detaching apparatus **200** (see FIGS. **1**A through **1**C) in that the gas supply unit **280** and a gas supply pathway **290** are provided. The gas supply pathway **290** is in communication with the gas pathways **12**.

In the substrate detaching apparatus **200**E, the control unit **240** may control the gas supply unit **280** to supply gas to the attraction surface **30***a* of the electrostatic chuck **30** at any selected timing. Specifically, the control unit **240** may control the gas supply unit **280** such that a gas is supplied to the attraction surface **30***a* via the gas supply pathway **290** and the gas pathways **12** while the moving unit **210** pushes the substrate **100**.

In this manner, it is preferable to provide the gas supply unit **280** that supplies gas to the attraction surface **30***a* via the gas pathways **12** of the electrostatic chuck **30** while the moving unit **210** pushes the substrate **100**. Provision of gas during the pushing of the substrate **100** by the moving unit **210** causes a force acting in the direction opposite a residual attraction force to be applied to the substrate **100**, thereby reducing the effect of the residual attraction force. As a result, the moving unit **210** can horizontally move the substrate **100** with a smaller force.

According to at least one embodiment, a substrate detaching apparatus is provided that is capable of reducing the effect of a residual attraction force of an electrostatic chuck.

All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiment(s) of the present inventions have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

What is claimed is:

1. A substrate detaching apparatus for detaching a substrate attracted to and held onto an attraction surface of an electrostatic chuck, comprising

a moving unit configured to push a side edge of the substrate to move the substrate while maintaining contact with the attraction surface in a direction parallel to the attraction surface, an upper surface of the substrate being free of physical contact while the side edge of the substrate is being pushed, the side edge of the substrate being pushed while a residual attraction force of the electrostatic chuck applied to the substrate substantially remains.

2. The substrate detaching apparatus as claimed in claim **1**, wherein the moving unit has two or more elongate parts spaced apart from each other, the elongate parts having a longitudinal direction thereof aligned in a direction of movement of the substrate.

3. The substrate detaching apparatus as claimed in claim **1**, wherein the moving unit has a curved part that has a same radius of curvature as the substrate and that is capable of contacting with less than or equal to half of a circumference of the substrate in a plan view.

4. The substrate detaching apparatus as claimed in claim **3**, wherein the curved part has a plurality of vacuum suction holes in a lateral surface toward the substrate.

5. The substrate detaching apparatus as claimed in claim **1**, further comprising guide rails spaced apart from each other and having a longitudinal direction thereof aligned in a direction of movement of the substrate, the guide rails guiding movement of the substrate from both sides.

6. The substrate detaching apparatus as claimed in claim **5**, wherein at least one of the guide rails has a plurality of vacuum suction holes in a lateral surface facing the substrate.

7. The substrate detaching apparatus as claimed in claim **1**, further comprising a loading arm situated opposite from the moving unit across the electrostatic chuck, the loading arm configured to carry the substrate detached from the attraction surface, wherein a loading surface of the loading arm is situated on a line on which a center of the substrate moves, and is coplanar with the attraction surface.

8. The substrate detaching apparatus as claimed in claim **1**, further comprising a gas supply unit configured to supply gas to the attraction surface via gas pathways of the electrostatic chuck while the moving unit pushes the substrate.

9. The substrate detaching apparatus as claimed in claim **1**, wherein a surface roughness Ra of the attraction surface is less than or equal to 0.2 μm.

10. The substrate detaching apparatus as claimed in claim **1**, wherein the attraction surface is constituted by top surfaces of a plurality of protrusions, and the top surfaces coming in contact with the substrate have edges with a rounded profile.

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